Rahul Yedida

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Employment

LexisNexis Legal & Professional — Raleigh, NC

Senior Data Scientist II — *Oct 2026 - Present*

- Fine-tuned small structured-decision models to match LLM-baseline performance (macro-F1 92.7%) at 177 ms median latency on-device, after benchmarking 15 models across 6 tasks.

Senior Data Scientist I — *May 2024 - Oct 2026*

- Technical lead for an agentic legal drafting system; led its migration from a RAG pipeline and own its ongoing development.
- On a 3-person team, increased the share of drafts rated suitable for users by attorney reviewers from 40% to 86% for complaints and 59.2% to 76% for motions.
- Reduced customer-facing product runtime by 24.8% through LLM call profiling and parallelization.
- Built a contrastive-learning filter using InfoNCE with alignment and uniformity losses to remove unrelated legal passages from search results; median filtering latency of 12 ms across 40 queries.
- Built team developer tooling: custom Python lint rules beyond `ruff`'s set, enforced in CI; a Rust MITM proxy with request replay that teammates use to debug without waiting on upstream responses; and a web UI for visualizing intermediate agent steps.
- Mentored an intern on a large-scale, full factorial experiment investigating Pareto-optimal configurations for draft quality and latency.
- Profiled high pod memory usage and introduced lazy module loading, reducing time at peak memory from 86.4% to 68% of runtime.
- **Technology:** Python, Litestar, Google ADK, React, Rust, TypeScript, Tailwind

North Carolina State University — Raleigh, NC

PhD Student — *Aug 2019 - Jul 2024*

- **State-of-the-art hyper-parameter optimization:** Derived loss-curvature estimates for smoothness- and flatness-guided search; the smoothness-based method matched or beat prior SOTA on 17 of 19 software datasets.
- **Better, faster deep learning for SE:** Introduced fuzzy oversampling and optimized neural-network hyper-parameters, surpassing prior SOTA with median relative gains of 15% in defect-prediction F1 and 12% in AUC across four code smells. Built tuned feedforward models that matched or beat SOTA across 15 issue-lifetime benchmarks in minutes instead of hours.
- **Microservice partitioning:** Developed a graph neural network partitioner with a custom weighted loss; reduced the share of cross-service calls by a median 66% versus prior SOTA across four applications.
- **Semi-supervised learning:** Combined oversampling, nearest-neighbor labeling, and hyper-parameter optimization for state-of-the-art static-analysis warning triage: median AUC 0.92 on eight Java projects using 10% of training labels.
- **Teaching:** TA for 830 students over five semesters in C programming and software engineering.

Relevant Projects

pi-permission-system (maintained fork) — May 2026 - Present

TypeScript, Node.js — *GitHub*

Maintain a permission-system fork for the Pi coding agent. Redesigned shell permissions using grammar-based analysis of pipelines, substitutions, and redirections, with protected-path checks and argument-aware approval rules. Extended enforcement to background commands and fixed concurrent subagent approval handling.

Q&A System for Zotero — Oct 2024 - Present

Rust, SQLite, LanceDB — *GitHub :: crates.io*

Built an academic question-answering and semantic search system over a Zotero library. Developed a Rust RAG library with configurable LLM, embedding, and reranking providers, tool calling, and limits on concurrent embedding requests. Wrote a custom PDF parser for academic papers (median throughput ≈ 2,400 pages/s across 11 PDFs; M4 Pro).

RAISE — Aug 2020 - Jun 2026

Python, Keras, PyTorch — *GitHub :: PyPI*

Sole developer of a Python package for reproducible ML experiments, providing data loading, evaluation metrics, statistical tests, and reference implementations of research algorithms. Uses Keras with a PyTorch backend; tested on Python 3.11-3.14 across Linux, Windows, and Apple Silicon macOS. Used by multiple research groups; downloaded 79k+ times on PyPI.

Flatness-guided HPO Nov 2023 - May 2026
Python, PyTorch, Keras *GitHub :: Paper*
Developed a hyper-parameter optimizer using closed-form loss-curvature estimates to select configurations with flatter losses. Implemented comparative benchmarks on MNIST, SVHN, OpenML, and Bayesmark.

GHOST Jun 2020 - Mar 2022
Python, Keras *GitHub :: Paper*
Developed the GHOST reference implementation for software defect prediction, combining adaptive oversampling with DODGE hyper-parameter optimization. Implemented configurable ablation experiments and Scott-Knott statistical comparisons using RAISE.

Skills

Languages: Rust, Python, TypeScript, Java, C++, Gleam

ML/AI: PyTorch, Keras, Hyper-parameter optimization, RAG, LLM evaluation, Agentic systems, Google ADK

Frameworks: React, FastAPI, Litestar, FastAPI

Databases: LanceDB, SQL, MongoDB, DynamoDB

Cloud: Google Cloud (Compute Engine, Cloud Storage, Vertex AI), AWS (EC2, S3, DynamoDB)

Education

North Carolina State University Raleigh, NC
Ph.D. Computer Science *Aug 2019 - Jul 2024*

- Advisor: *Dr. Tim Menzies*
- Dissertation: *Guidelines for the Application of Neural Technologies in Software Analytics (or: How to Do More with Less in SE)*

PES University Bangalore, India
B.E. Computer Science – Advisor: Dr. Snehanshu Saha *Aug 2015 - May 2019*

Honors

Distinguished Reviewer, ICSE 2026 Mar 2026
Google Developer Expert - ML and GCP Mar 2025 - Present
Google Cloud Champion Innovator - Cloud AI/ML Oct 2022 - Mar 2025
Google Cloud Research Innovator Feb 2022

Recent Publications

See full list on Google Scholar.

1. Nagarsekar, A., Bhat, M. P. A., Nesarkar, A., Godhwani, V., **Yedida, R.**, Challa, A., Danda, S., & Saha, S. (2026). A Generalisation Signal Need Not Be a Model-Selection Signal. In *I Can't Believe It's Not Better: Failure Modes of AI in Biology Workshop (NeurIPS 2026)*.
2. **Yedida, R.**, & Menzies, T. (2025). Is Hyper-Parameter Optimization Different for Software Analytics? *IEEE Transactions on Software Engineering, 51(6)*.
3. Baldassarre, M. T., Ernst, N., Hermann, B., Menzies, T., & **Yedida, R.** (2023). (Re)use of Research Results (is Rampant). *Communications of the ACM, 66(2), 75-81*.
4. **Yedida, R.**, Kang, H. J., Tu, K., Lo, D., & Menzies, T. (2023). How to Find Actionable Static Analysis Warnings: A Case Study with FindBugs. *IEEE Transactions on Software Engineering, 49(4), 2856-2872*.
5. **Yedida, R.**, Krishna, R., Kalia, A., Menzies, T., Xiao, J., & Vukovic, M. (2023). An Expert System for Redesigning Software for Cloud Applications. *Expert Systems with Applications, 219, 119673*.